CLIL UNIT 1 • NATURAL SCIENCE & ENGLISH INTEGRATION

WE ARE HUMANS & ADVENTURE HEROES!

Human Cells, Body Systems, Extreme Sports, Grammar & Prepositions

(Fusión didáctica: Biología humana, deportes de aventura, gramática y preposiciones)

1. SCIENCE
Cells & Levels

2. SPORTS & GEAR
Extreme Gear

3. GRAMMAR LAB
Present & Adverbs

4. PREPOSITIONS
Hero Locations



1. The Human Machine in Motion!

¿Qué ocurre en nuestro cuerpo cuando corremos, escalamos o nadamos?

💓 Circulatory Pump

Aparato circulatorio en acción

Your heart beats faster to pump oxygenated blood to working muscles.
English: *"My heart is beating at 140 beats per minute!"*

🫁 Oxygen Delivery

Aparato respiratorio

Lungs take in oxygen ($ ext{O}_2$) and expel carbon dioxide ($ ext{CO}_2$) to fuel endurance.
English: *"Athletes breathe deeply to get more oxygen."*

⚡ Cell Mitochondria

Centrales energéticas celulares

Millions of muscle cells burn glucose and oxygen to generate explosive ATP kinetic power!
English: *"Cells produce the energy to climb."*

🧠 Nervous Commands

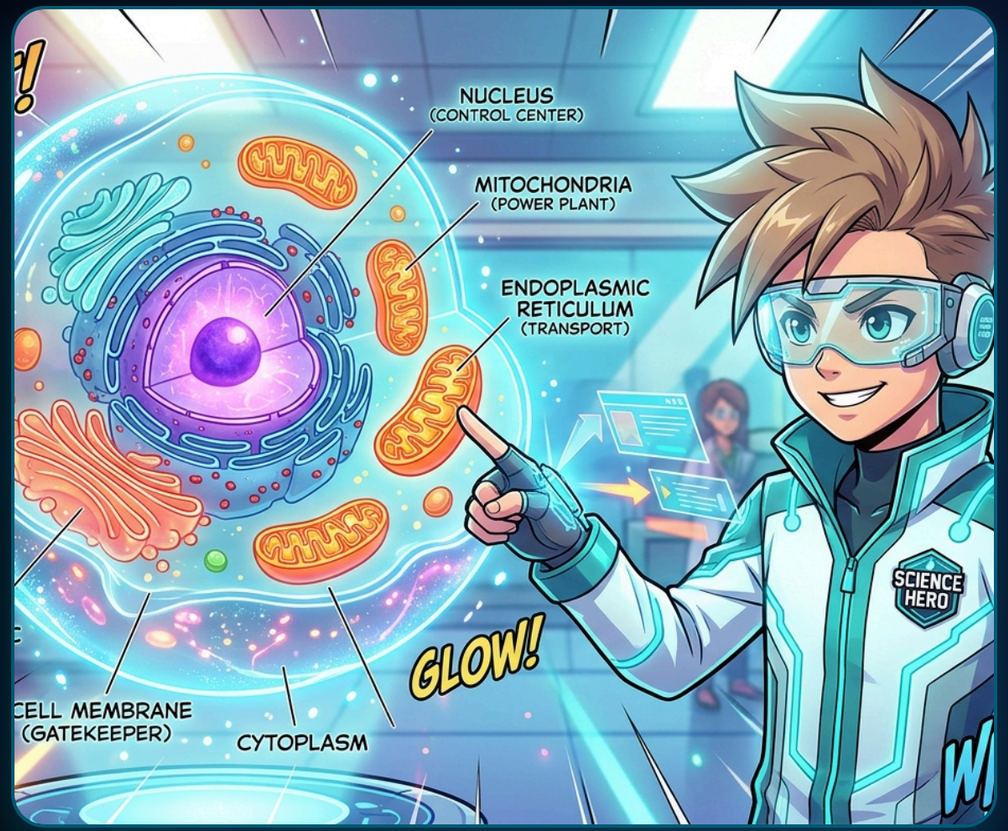
Sistema nervioso y reflejos

Your brain and nerves send electric impulses at 400 km/h to coordinate balance and agility.
English: *"The climber reacts with fast reflexes."*



2. Inside the Animal Cell

Anatomía celular: los 5 orgánulos clave de la célula animal que nos dan vida



1. **Cell Membrane** (Membrana Celular)
   The **protective border** that controls what enters (nutrients) and leaves (wastes).

2. **Nucleus** (Núcleo)
   The **command centre** housing DNA (genetic instructions) for all cell activities.

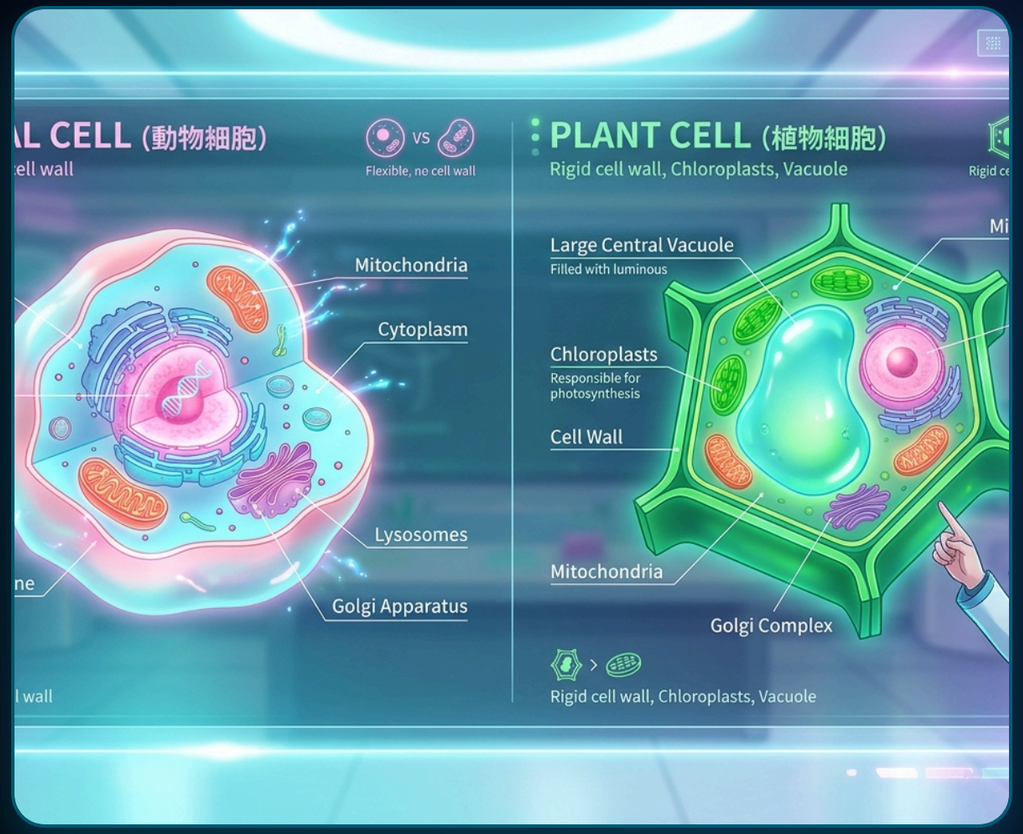
3. **Cytoplasm** (Citoplasma)
   Clear, jelly-like fluid where organelles float and vital chemical reactions occur.

4. **Mitochondria** (Mitocondrias)
   The **power plants**! They transform nutrients and $ ext{O}_2$ into muscular energy.

5. **Vacuoles & Organelles** (Vacuolas y orgánulos)
   Storage bubbles that store water, proteins, and minerals until needed.

3. Animal Cell vs. Plant Cell

¿Por qué los humanos podemos movernos y las plantas son rígidas y hacen fotosíntesis?



Animal Cell (Humans & Animals)

FLEXIBLE & ACTIVE

- **Membrane, nucleus, cytoplasm, mitochondria.**
- **No cell wall:** Allows animal cells to be flexible so we can bend joints, sprint, and jump!
- Contains small, multiple vacuoles.

Plant Cell (Exclusive Superhero Gear)

RIGID & GREEN

1. **Cell Wall (Pared celular):** Rigid shield made of cellulose that holds trees tall.
2. **Chloroplasts (Cloroplastos):** Solar factories filled with chlorophyll for photosynthesis.
3. **Giant Vacuole (Gran vacuola):** Stores water to keep plant cells firm.

English Grammar Link: *"Humans haven't got chloroplasts, so we must eat food to obtain energy!"*

4. The 5 Levels of Organisation

De la célula al organismo completo: la jerarquía biológica que forma a un deportista

1

Cell
CÉLULA

The fundamental living unit.

Example:
Muscle Cell
(Célula muscular)

2

Tissue
TEJIDO

Group of identical cells working as one.

Example:
Muscle Tissue
(Tejido muscular)

3

Organ
ÓRGANO

Tissues united for a specific mission.

Example:
The Heart
(El corazón)

4

System
APARATO / SISTEMA

Organs working together in synergy.

Example:
Circulatory System
(Aparato circulatorio)

5

Organism
ORGANISMO

The complete living adventure hero!

Example:
Human Climber
(Ser humano)

5. Adventure Sports • Deportes de Aventura

Vocabulario de acción: deportes de aventura y los verbos para practicarlos



Rock climbing

ESCALADA EN ROCA

Climbing up natural rock formations or artificial walls using ropes.

👉 *We go rock climbing at weekends.*



Kayaking

PIRAGÜISMO / KAYAK

Navigating rivers and rapids in a small canoe using a double paddle.

👉 *They go kayaking down rapid rivers.*



Scuba diving

SUBMARINISMO / BUCEO

Swimming underwater using an oxygen tank and breathing regulator.

👉 *She goes scuba diving to see coral reefs.*



Snowboarding

SNOWBOARD / TABLA DE NIEVE

Gliding down snow-covered slopes on a single wide board attached to boots.

👉 *I go snowboarding in the mountains.*



Surfing

SURF

Riding ocean waves standing up on a surfboard towards the shore.

👉 *Do you go surfing in summer?*



Caving

ESPELEOLOGÍA

Exploring underground caves and subterranean caverns with headlamps.

👉 *Explorers go caving with safety helmets.*

6. Safety Gear & Human Anatomy

El equipo de aventura como armadura biológica que protege nuestros órganos y huesos

Helmet (Casco)

Protects: The **Cranium (Skull)** & the **Brain**.

Prevents concussions and protects the central nervous system from rockfalls.

Wetsuit (Traje de neopreno)

Protects: The **Skin** & **Body Temperature**.

Prevents hypothermia by trapping a thin layer of warm water against your skin.

Harness & Ropes (Arnés y cuerdas)

Protects: The **Spine (Vertebrae)** & **Pelvis**.

Distributes fall forces safely across strong hip bones rather than fragile joints.

Life Jacket (Chaleco salvavidas)

Protects: The **Respiratory System**.

Keeps the mouth and nose above water so lungs can continue taking in oxygen!



7. Present Simple vs. Present Continuous

Diferenciar hábitos y hechos científicos de acciones que ocurren en este preciso instante

Present Simple

HABITS & FACTS

Used for regular habits, sports routines, and scientific laws:

- *The heart **pumps** blood every second.*
- *Alex **goes** kayaking on Saturdays.*
- *They **do not climb** without ropes.*

🔑 Clue words: ***always, usually, every week, on Mondays, never.***

Present Continuous

HAPPENING RIGHT NOW

Used for actions happening at this exact moment:

- *Look! Maya **is climbing** the mountain now.*
- *His muscle cells **are burning** ATP energy!*
- *We **are wearing** our life jackets.*

🔑 Clue words: ***now, right now, at the moment, Look!, Listen!***

8. Adverbs of Frequency • ¿Con qué frecuencia?

La escala porcentual de frecuencia y su posición exacta en la frase en inglés

100%	80%	60%	40%	15%	0%
Always	**Usually**	**Often**	**Sometimes**	**Rarely**	**Never**
Siempre	Normalmente	A menudo	A veces	Raras veces	Nunca

📌 Rule 1: Before Normal Verbs

[Subject] + [Adverb] + [Main Verb]

- *Humans **always need** clean water.*
- *He **usually wears** a helmet when climbing.*
- *They **never go** caving without torches.*

📌 Rule 2: After the Verb TO BE

[Subject] + [am/is/are] + [Adverb]

- *Athletes **are always** careful with safety.*
- *The weather in mountains **is often** unpredictable.*
- *She **is never** afraid of heights.*

9. Superheroes & Prepositions of Place

Preposiciones de lugar y movimiento con cubos, plataformas, balones y héroes



ON

Encima / Sobre la superficie

- *The hero is **ON** the cube with the football.*
- *Anatomy: The skull rests **ON** the spine.*

UNDER

Debajo de

- *The hero is crouching **UNDER** the platform.*
- *Sports: The diver swims **UNDER** the water.*

IN / INSIDE

Dentro de

- *The hero is **IN** the glowing glass cube.*
- *Science: Organelles float **IN** the cytoplasm!*

NEXT TO

Al lado de / Junto a

- *The hero is standing **NEXT TO** the basketball.*
- *Gear: The kayak is **NEXT TO** the river.*

OVER & THROUGH (Movement)

Por encima de & A través de

- *The athlete jumps **OVER** the high hurdle.*
- *Blood cells travel **THROUGH** our veins and arteries!*

10. Stages of Human Life

El ciclo vital humano: desarrollo celular, físico y capacidades deportivas



Infancy

0 - 3 AÑOS

Rapid cell division, learning to crawl, walk, and speak first words.



Childhood

3 - 11 AÑOS (YOU!)

Learning adventure sports, agility, team coordination, brain development.



Adolescence

11 - 18 AÑOS

Puberty, rapid growth spurts, muscle mass gains, hormonal maturation.



Adulthood

18 - 65 AÑOS

Peak physical performance, endurance, careers, independence, and family.



Old Age

65+ AÑOS

Retirement and wisdom. Bones need care, gentle walking, and healthy habits.

11. Vocabulary Vault • Banco Bilingüe

Términos esenciales en inglés con su traducción en castellano y definición concisa

Cell
CÉLULA
Smallest unit of living organisms.

Mitochondria
MITOCONDRIA
Powerhouse producing muscular energy.

Tissue
TEJIDO
A group of specialized cells.

Organ
ÓRGANO
Tissues working together (e.g. heart).

Rock Climbing
ESCALADA EN ROCA
Ascending mountain rock walls.

Kayaking
PIRAGÜISMO / KAYAK
Paddling through river rapids.

Helmet
CASCO DE SEGURIDAD
Gear protecting skull and brain.

Harness
ARNÉS DE ESCALADA
Safety straps protecting the spine.

Wetsuit
TRAJE DE NEOPRENO
Insulating suit maintaining warmth.

Life Jacket
CHALECO SALVAVIDAS
Buoyancy aid keeping head afloat.

Bipedalism
BIPEDESTACIÓN
Walking upright on two feet.

Opposable Thumbs
PULGARES OPONIBLES
Thumbs enabling tool grip.

Always (100%)
SIEMPRE
At all times without exception.

Usually (80%)
NORMALMENTE
Most of the time in routines.

Never (0%)
NUNCA
Not at any time.

Right Now
AHORA MISMO
At this exact present moment.

12. Speaking Arena • Diálogo de Clase

Práctica oral por parejas: una entrevista entre un científico deportivo y un atleta héroe

DR. NEXUS: "Hello, Leo! What sport **are you practicing right now**?"

LEO (HERO): "I **am climbing** the steep canyon wall! I **usually climb** three times a week."

DR. NEXUS: "Impressive! **How are your cells and muscles feeling**?"

LEO (HERO): "My **mitochondria are producing** lots of energy, but my heart **is beating** very fast!"

DR. NEXUS: "Good! Remember: heroes **always wear** a helmet **ON** their head and a harness **AROUND** their waist!"

13. CLIL Hero Challenge!

4 preguntas interactivas para poner a prueba conocimientos de Ciencias e Inglés

🔬 Q1 • Science

Which organelle acts as the cellular power plant producing energy for muscles?

ANSWER:
Mitochondria! ⚡

📝 Q2 • Grammar

Choose the correct sentence for right now:
A) She climbs now.
B) She is climbing now.

ANSWER:
B) She is climbing! 🧗

⚽ Q3 • Prepositions

Where does a climber wear a helmet?
A) Under the feet
B) On the head

ANSWER:
B) ON the head! ⛑️

🛡️ Q4 • Gear

What suit protects your skin and temperature during surfing or scuba diving?

ANSWER:
The Wetsuit! 🏄

🎉 MISSION ACCOMPLISHED!

You are now equipped with the scientific knowledge of the human body and the English language power to conquer any adventure!

🔬 **Science Core:**
Cells, organelles, 5 levels of organisation, vital functions, and life stages.

🗣️ **English Core:**
Adventure sports, protective gear, present continuous, and frequency adverbs.

